SkillLink — Service Marketplace System

Deliverable 1 · Sprint 1 Documentation

Module	User Identity & Profile Management
Sprint Goal	Establish foundational architecture for user onboarding and role-based profile management
Methodology	Scrum (Iterative & Incremental)
Team	Zaki Nabeel (Team Lead) · Shehryar Faisal (SCRUM Master) · Muhammad Sameer (Developer)

1. Introduction

SkillLink is a digital service marketplace platform connecting clients who need skilled professionals with service providers across trades and expertise. It addresses the absence of a trustworthy, centralised portal in Pakistan where clients can discover, vet, and book providers such as plumbers, electricians, graphic designers, tutors, and software developers.

2. Project Vision

Vision: To become the leading digital service marketplace in Pakistan, empowering clients to access quality services instantly while enabling skilled providers to grow their businesses through a transparent, secure, and user-friendly platform.

- Seamless discovery of providers across 20+ skill categories.
- End-to-end service delivery — from booking to payment to review.
- Secure, role-based environment for clients and providers.
- Trust through transparent profiles, reviews, and ratings.

3. Intended Use of the System — Stakeholders

SkillLink serves **Clients** (seek services), **Service Providers** (offer expertise), and a **Platform Admin** (moderation). A Dev Team maintains the system.

Stakeholder	Role	Needs
Clients	End users seeking services	Easy search, booking, secure payments, provider reviews
Service Providers	Professionals offering skills	Profile visibility, booking management, earnings tracking
Platform Admin	System moderator	User management, dispute resolution, platform analytics
Dev Team	Builders & maintainers	Clear specs, testable stories, defined acceptance criteria

4. Features & Overall Functionality

Identity & Profile Mgmt	Role-based registration, JWT login, provider profiles, portfolio links, contact management.
Search & Discovery	Category search, geo-filter, skill taxonomy tagging, provider profile view.
Booking & Scheduling	Service requests, provider accept/decline, booking history.
Reviews & Ratings	Post-job reviews, provider responses, aggregate ratings.
Messaging & Payments	In-app chat, secure payment gateway, earnings dashboard.
Admin & Moderation	User management, reported content review, analytics dashboard.

5. User Stories — Sprint 1

US-01: Account Registration

***As a** new user, **I want to** create an account, **so that** I can access the SkillLink marketplace as either a Client or a Service Provider.*

Sub-Stories:

- 1.1 System shall validate email format and check for existing accounts.
- 1.2 System shall enforce a strong password policy (min 8 chars, 1 number, 1 special char).
- 1.3 User must select a role (Client or Provider) during signup.

Given	When	Then
User is on the signup page	They submit valid email, password, and role	Account is created and a verification email is sent
Email already exists in DB	User clicks Submit	System displays error prompting login

US-02: Secure Authentication

***As a** registered user, **I want to** log in securely, **so that** I can access my personalized dashboard and manage my bookings.*

Sub-Stories:	Given	When	Then
• 2.1 Implement Forgot Password flow via secure email link.	User is on the login page	They enter correct credentials	Redirected to their role-based dashboard
• 2.2 Implement session persistence using JWT for Remember Me functionality.	User clicks Forgot Password	They enter their registered email	A time-sensitive password reset link is sent

US-03: Provider Profile Setup

***As a** Service Provider, **I want to** set up my professional profile, **so that** clients can view my expertise, portfolio, and contact me for jobs.*

Sub-Stories:	Given	When	Then
• 3.1 Upload and crop a professional profile picture. • 3.2 Select primary and secondary skill categories from a predefined taxonomy. • 3.3 Write a formatted professional bio.	Provider is in profile settings	They upload a JPG/PNG under 5 MB	Image is saved and displayed as avatar
	Provider leaves Skill Category blank	They attempt to save	System prevents saving and highlights required field

US-04: Client Contact Management

***As a** Client, **I want to** manage my contact information and location, **so that** providers can accurately reach me to fulfil service requests.*

Sub-Stories:	Given	When	Then
• 4.1 Update phone number with regional validation. • 4.2 Set physical location/address for geographic service filtering.	Client updates phone number	They enter a valid 11-digit number	System saves the update successfully

6. Structured Specifications (US-03: Provider Profile Setup)

Spec ID	Spec-03.1
Description	Upload, validate, and store a profile avatar.
Pre-cond	Logged in as Service Provider.
Inputs	.jpg, .jpeg, .png image file.
Process	1. Check file format. 2. Check size vs 5 MB limit. 3. Compress and store in cloud bucket. 4. Update DB record with image URL.
Outputs	Updated avatar confirmed on UI.
Exceptions	FileTooLarge (>5 MB). UnsupportedMediaType (bad format).

Spec ID	Spec-03.2
Description	Tag profile with searchable service categories.
Pre-cond	Profile page loaded with active DB connection.
Inputs	Array of category IDs e.g. [ID_Plumbing].
Process	1. User selects 1-5 skills from dropdown. 2. System maps IDs to user foreign key in DB.
Outputs	Profile metadata updated; appears in relevant search results.
Exceptions	0 skills: UI validation error. >5: disable further selection.

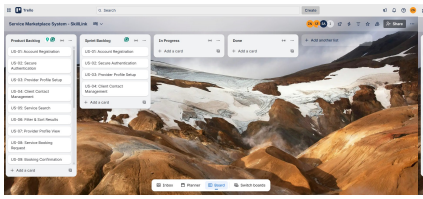
Spec ID	Spec-03.3
Description	Text area for providers to pitch services to clients.
Pre-cond	User is on the profile edit screen.
Inputs	Alphanumeric text string.
Process	1. Real-time character counter. 2. Sanitise input (XSS prevention). 3. Save text to DB record.
Outputs	Formatted text rendered on public provider landing page.
Exceptions	Length < 50 chars: show Bio too short warning, disable save.

Spec ID	Spec-03.4
Description	Validate and store external portfolio URLs.
Pre-cond	User is on the profile edit screen.
Inputs	URL string (GitHub, Behance, LinkedIn, etc.).
Process	1. Regex validates http/https format. 2. Store verified link in profile table.
Outputs	Clickable portfolio icons/links on public profile.
Exceptions	Invalid URL: highlight input red, prompt for correction.

7. Scrum Board — Sprint 1 (Trello)

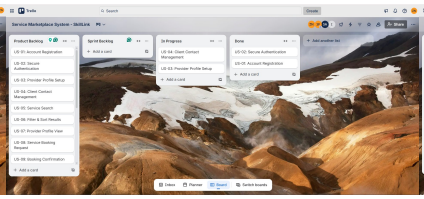
Three Trello snapshots document Sprint 1 progression across Product Backlog, Sprint Backlog, In Progress, Done, and Leftover List columns.

Snapshot 1 — Start of Sprint



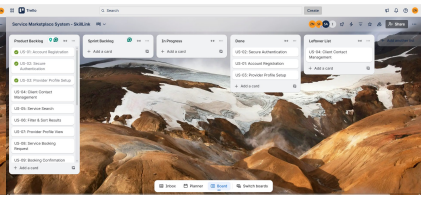
All stories in Sprint Backlog. Board empty.

Snapshot 2 — Mid Sprint



US-01 & US-02 Done. US-03 & US-04 In Progress.

Snapshot 3 — End of Sprint



US-01/02/03 Done. US-04 in Leftover List.

Sprint 1 User Story Summary

User Story	Sprint 1 Status	Final Column	Notes
US-01: Account Registration	Done	Done	Completed within Sprint 1
US-02: Secure Authentication	Done	Done	Completed within Sprint 1
US-03: Provider Profile Setup	Done	Done	Completed within Sprint 1
US-04: Client Contact Management	Leftover	Leftover List	Carried to Sprint 2

8. Non-Functional Requirements (NFR) — Natural Language Specification

NFRs define quality attributes the system must satisfy. Format: *The system shall [condition] [measurable criterion].*

#	Category	NFR Statement	Priority
1	**Performance**	The system shall load any page within 3 seconds for up to 500 concurrent users.	**High**
2	**Security**	All passwords shall be hashed using bcrypt; no plain-text credentials shall be stored.	**Critical**
3	**Security**	All API communication shall use HTTPS/TLS 1.2 or higher to prevent interception.	**Critical**
4	**Availability**	The platform shall maintain 99.5% uptime; maintenance communicated 24 hrs in advance.	**High**
5	**Scalability**	The architecture shall support horizontal scaling for a 10x increase in registrations.	**Medium**
6	**Usability**	The interface shall conform to WCAG 2.1 AA accessibility standards.	**High**
7	**Reliability**	User data shall be auto-backed up every 24 hours with 4-hour restore capability.	**High**
8	**Maintainability**	Source code shall follow coding standards, documented for 2-day developer onboarding.	**Medium**

9. Iteration 1 — Implementation Screenshots

Trello board snapshots in Section 7 serve as Iteration 1 implementation evidence. Application UI screenshots will be added in subsequent deliverables as modules are deployed.

10. Work Division

Sprint 1 responsibilities were distributed across three team members — all 6th-semester BS CS students at FAST NUCES Islamabad — based on individual core competencies.

Name & GitHub	Role(s)	Bio	Sprint 1 Responsibilities
Zaki Nabeel Team Lead GitHub: ZakiNabeel	Project Manager Tester	Tech with Purpose. 6th-sem BS CS, FAST NUCES Islamabad. Industrial experience at OGDCL. Passionate about code, design, and leading teams through challenges.	Sprint planning, milestone tracking, test case design, QA validation, stakeholder communication
Shehryar Faisal GitHub: Shehryarf67	SCRUM Master Analyst / Architect	Loves building innovative products. Deep skill set in web dev and team management. Designs applications and leads teams from the front to the desired outcome.	Sprint facilitation, backlog grooming, system architecture, structured specs, Trello board management

Muhammad Sameer GitHub: —	Developer UI Designer	Keen eye for aesthetics, strong software foundation. Bridges design and functionality through intuitive UIs and clean code. Transforms complex ideas into seamless experiences.	Front-end implementation, UI/UX design, React components, profile setup screens, responsive layout

Sprint 1 Outcome: 3 of 4 planned user stories completed. US-04 (Client Contact Management) carried forward to Sprint 2 as leftover. Primary sprint goal — foundational user onboarding and role-based profile management — substantially achieved.